

Abhinav Vummidichetty

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TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, Bash

Big Data & Streaming: Apache Kafka, Apache Spark (PySpark), Apache Hadoop, Apache Airflow, dbt

Infrastructure & Cloud: GCP (Dataflow), Snowflake, AWS, Docker, PostgreSQL, Git

Applied Methodologies: ETL/ELT Pipelines, Data Modeling, Data Warehousing, Real-Time Data Processing

EDUCATION

San Jose State University — *M.S. in Data Analytics*

Dec 2026

Coursework: Machine Learning, Statistical Modeling, Deep Learning, Big Data Analytics

Andhra University College of Engineering — *B.Tech in Computer Science*

May 2024

EXPERIENCE

Beet (gobeet.app)

Remote

Data Engineering Volunteer

Jan 2026 – Present

- Built **ETL/ELT** workflows in **Python** and **SQL** to ingest event data from 5+ sources into a centralized warehouse, enforcing schema validation and **data integrity** checks to prevent silent data corruption downstream.
- Designed storage schemas and partitioning strategies to minimize full-table scans; collaborated with product, engineering, and data science teams to translate business requirements into structured pipeline specifications.
- Maintained documentation for all data assets and pipeline decisions, establishing **data governance** conventions that improved cross-team reproducibility and reduced analyst onboarding time.

PROJECTS

TaxiFlow: Real-Time Streaming Pipeline — Python, Apache Kafka, Docker, PostgreSQL

- Designed a **real-time data processing** pipeline using **Apache Kafka** — engineering producer, processor, and consumer scripts to fully decouple ingestion from transformation, enabling independent scaling of each layer across millions of NYC taxi trip records.
- Implemented multi-stage data validation logic at the consumer level, achieving 95%+ accuracy across millions of events while reducing end-to-end message latency by 30% through partition tuning and batched offset commits.

MarketFlow: Automated ETL Stock Pipeline — Python, Apache Airflow, dbt, SQL

- Architected an automated **ETL/ELT** pipeline using **Apache Airflow** for DAG-based orchestration and **dbt** for structured **SQL** transformations, eliminating 45% of manual processing tasks and reducing runtimes by 30% via incremental model execution.
- Embedded data quality tests into dbt models to validate accuracy, uniqueness, and referential integrity; configured Airflow DAG retry logic and SLA alerts to ensure business-critical pipelines met latency requirements.

Spark Volatility: Large-Scale Analytics with PySpark — Python, PySpark, Apache Hadoop

- Built a **PySpark** analytics solution on **Apache Hadoop** using MapReduce to compute annualized volatility and daily return metrics across millions of records — chose Spark over Pandas to avoid single-node memory ceiling limitations.
- Tuned Spark transformations including partition coalescing and broadcast joins to reduce processing time by over 40% while achieving 95% accuracy in volatility computation validated against benchmark datasets.

VespaSearch: Scalable Semantic Recommendation Engine — Python, Vespa, GCP

- Built a semantic search and recommendation engine using **Vespa's** distributed indexing — chose Vespa over Elasticsearch for native approximate nearest-neighbor search combined with structured query evaluation in a single retrieval pass.
- Tuned ranking profiles and relevance signals to achieve 90% retrieval precision and 25% lower query latency; boosted data indexing throughput by 40% through parallel feed workers and schema-level compression.